

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

Submitted solver team

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Abstract

We present `cf2u`, an original cell-centered unstructured finite-volume solver for the two-dimensional compressible Navier–Stokes equations of a calorically perfect gas, parallelized with MPI domain decomposition and METIS graph partitioning. The solver uses second-order piecewise-linear reconstruction with a smooth Venkatakrishnan limiter, a Rusanov (local Lax–Friedrichs) approximate Riemann solver, Newtonian/Fourier viscous fluxes, and an implicit LU-SGS/SGS pseudo-time march with a force-plateau convergence criterion; the unsteady cylinder case uses a true BDF2 dual-time formulation. All eight required cases were run to converged or statistically periodic states on the supplied meshes. We report residual and force histories, force coefficients, surface distributions, field contours, a parallel scaling study up to eight ranks, and an honest account of limitations.

1 Introduction

The objective of this benchmark is to design, implement, and validate a complete CFD solver rather than a prototype: a two-dimensional unstructured finite-volume code for the compressible Navier–Stokes equations, executed with MPI domain decomposition, and applied to a family of NACA0012 airfoil and circular-cylinder cases spanning inviscid and laminar flow from subsonic to supersonic Mach number, including an unsteady laminar vortex street.

The eight required cases are: three inviscid NACA0012 cases (Mach 0.15, 0.8, 2.0), three laminar NACA0012 cases at Reynolds number 5000 (Mach 0.15, 0.8, 2.0), a steady laminar cylinder case at Reynolds number 20, and an unsteady laminar cylinder case at Reynolds number 200 that must be advanced to a statistically periodic vortex-shedding state. This report connects our numerical choices to the observed results, states convergence status honestly, and does not present any failed or partial run as successful. All eight cases completed successfully; the convergence evidence is given in section 9.

2 Governing Equations and Nondimensionalization

We solve the two-dimensional compressible Navier–Stokes equations in conservative form for the state

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T,$$
$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

The gas is calorically perfect with ratio of specific heats $\gamma = 1.4$ and gas constant $R = 1.0$:

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho}.$$

For viscous cases the viscous fluxes use a Newtonian stress tensor and Fourier heat conduction,

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x),$$

$$\mathbf{q} = -k\nabla T, \quad k = \frac{\mu c_p}{Pr},$$

with Prandtl number $Pr = 0.72$ and specific heat $c_p = \gamma R / (\gamma - 1)$. A constant dynamic viscosity is used, chosen to match the case Reynolds number,

$$\mu = \frac{\rho_\infty U_\infty L_{ref}}{Re}.$$

The cases are run in dimensional-like consistent units with reference density $\rho_\infty = 1$, reference velocity $U_\infty = 1$, reference length $L_{ref} = 1$, and freestream pressure set by the target Mach number $p_\infty = \rho_\infty U_\infty^2 / (\gamma M_\infty^2)$. Force coefficients use the dynamic pressure $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$, the reference area $A_{ref} = 1$, and reference length $L_{ref} = 1$:

$$C_D = \frac{F_x^{body}}{q_\infty A_{ref}}, \quad C_L = \frac{F_y^{body}}{q_\infty A_{ref}}, \quad C_m = \frac{M_z}{q_\infty A_{ref} L_{ref}},$$

resolved into pressure and viscous parts. Because $U_\infty = 1$ and $\rho_\infty = 1$, the dynamic pressure is $q_\infty = 0.5$.

3 Meshes, Boundary Tags, and Case Inputs

Both meshes are read from CGNS files with the CGNS/HDF5 mid-level library. The reader extracts the node coordinates, the element connectivity (triangles and quadrilaterals), the boundary-condition point ranges that reference boundary edge elements, and the zone-to-zone **Abutting1to1** connectivity used to stitch the two cylinder zones. From the connectivity we build the cell-face dual graph and the geometric quantities: cell volumes from the polygon area, face normals as the outward area vector of each edge (rotated 90° from the edge tangent with a consistent left-to-right orientation), and boundary-face lengths. Boundary families are mapped to physical conditions through the case **boundary_conditions** block: farfield, inviscid slip wall, or viscous no-slip adiabatic wall. Every family named in the case file must exist in the mesh and vice versa, which is verified at start-up.

Table 1 summarizes the meshes and Table 2 the case parameters. The NACA0012 mesh mixes triangles and quadrilaterals with an extreme cell-volume range (roughly 1.5×10^{-9} to 28) because of the boundary-layer stretching; the cylinder mesh is split into a near-body zone and a far zone joined by conforming interfaces.

Table 1: Mesh summary.

Mesh	Cells	Cell types	Boundary families	Notes
NACA0012_H2.cgns	20816	tri + quad	farfield, wall	single zone
CylinderB1.cgns	10185	tri + quad	wall, far	two zones, stitched

Table 2: Case parameters.

Case	M_∞	Re	Mode	Target (orders)	max steps	CFL
naca0012_m015_inviscid	0.15	–	inviscid	4	20000	$1 \rightarrow 100$
naca0012_m080_inviscid	0.80	–	inviscid	4	30000	$1 \rightarrow 100$
naca0012_m200_inviscid	2.00	–	inviscid	3	40000	$0.2 \rightarrow 50$
naca0012_m015_laminar_re5000	0.15	5000	laminar	4	40000	$1 \rightarrow 100$
naca0012_m080_laminar_re5000	0.80	5000	laminar	4	40000	$0.5 \rightarrow 80$
naca0012_m200_laminar_re5000	2.00	5000	laminar	3	50000	$0.2 \rightarrow 50$
cylinder_m010_laminar_re20	0.10	20	laminar	5	30000	$1 \rightarrow 100$
cylinder_m010_laminar_re200	0.10	200	laminar	–	30000 steps	1.0 (fixed)

4 Spatial Discretization

The conservative state is stored at cell centers. The semi-discrete residual for cell i is the sum of numerical fluxes over its faces,

$$R_i = \sum_{f \in \partial i} \left(\hat{\mathbf{F}}_f^i - \hat{\mathbf{F}}_f^v \right),$$

where each face normal $\mathbf{n}_f$ is the outward area vector oriented from the left cell to the right cell (or out of the domain at a boundary). Interior faces contribute a flux computed from the left and right reconstructed states; boundary faces contribute a flux computed from the interior reconstructed state and the boundary condition, as detailed in section 5.

4.1 Second-Order Reconstruction

Cell gradients of the primitive variables $\mathbf{W} = (\rho, u, v, p, T)$ are computed with an inverse-distance-weighted least-squares (LSQ) fit. For each cell the stencil contains all face-neighbors plus, for cells touching a boundary, a boundary point at each boundary-face center whose value is supplied by the boundary condition (zero normal gradient for scalars, the tangential velocity at slip walls, zero velocity at no-slip walls, and the freestream at the farfield). The 2×2 LSQ system is weighted by $1/|\Delta \mathbf{x}|^2$ and solved exactly; cells with a singular or under-sized stencil fall back to a zero gradient (first order), which occurs only for isolated degenerate cells. Piecewise-linear reconstruction to the face centroid, $\mathbf{W}_f = \mathbf{W}_c + \phi (\nabla \mathbf{W}_c) \cdot (\mathbf{x}_f - \mathbf{x}_c)$, gives the left/right face states, with ϕ the limiter of section 4.2. Reconstruction is active in all production runs (spatial order 2); the only first-order path is the rare degenerate-stencil fallback and the optional positivity clamp.

4.2 Limiter and Positivity Control

Production runs use the smooth Venkatakrishnan limiter. For each cell and each limited variable, let $\Delta = \nabla \mathbf{W}_c \cdot (\mathbf{x}_f - \mathbf{x}_c)$ be the unlimited face increment and D the admissible increment toward

the neighbor extremum ($\mathbf{W}_i^{max} - \mathbf{W}_c$ if $\Delta > 0$, else $\mathbf{W}_i^{min} - \mathbf{W}_c$). The smooth limiter value is

$$\psi = \frac{1}{\Delta} \frac{(D^2 + \varepsilon^2) \Delta + 2\Delta^2 D}{D^2 + 2\Delta^2 + D\Delta + \varepsilon^2}, \quad \phi = \min_f \psi,$$

with $\varepsilon^2 = (Kh)^3$, h the cell length scale, and $K = 1$. Because the limiter is differentiable it avoids the limit-cycle stall that the non-smooth Barth–Jespersen limiter produced in our steady low-Mach computations; a Barth–Jespersen limiter is also implemented and selectable. Positivity is protected in two places: reconstructed face states are floored to $\rho, p \geq 10^{-8}$, and the conservative update is scaled back per cell until the updated density and pressure are positive.

4.3 Inviscid and Viscous Fluxes

The inviscid numerical flux is the Rusanov (local Lax–Friedrichs) flux

$$\hat{\mathbf{F}}^i = \frac{1}{2}(\mathbf{F}^i(\mathbf{W}_L) + \mathbf{F}^i(\mathbf{W}_R)) \cdot \mathbf{n} - \frac{1}{2} s_{max} (\mathbf{U}_R - \mathbf{U}_L) |\mathbf{n}|, \quad s_{max} = \max_{L,R} (|u_n| + a),$$

with dissipation scale 1.0. No entropy fix is required for the Rusanov flux. Viscous fluxes are evaluated from face gradients of velocity and temperature. Interior-face gradients use the corrected average of the two cell gradients,

$$\nabla \phi_f = \frac{1}{2}(\nabla \phi_L + \nabla \phi_R) + \mathbf{e} \left(\frac{\phi_R - \phi_L}{|\mathbf{x}_R - \mathbf{x}_L|} - \frac{1}{2}(\nabla \phi_L + \nabla \phi_R) \cdot \mathbf{e} \right),$$

which restores consistency on skewed cells. The viscous stresses and heat flux are assembled into $\hat{\mathbf{F}}^v$ and subtracted from the inviscid flux. The pressure force on a body is integrated separately from the tangential skin-friction force so that pressure drag/lift and viscous drag/lift are reported independently.

5 Boundary Conditions

Three boundary conditions are required by the cases.

Farfield. A characteristic (Riemann-invariant) farfield is the default. For a subsonic boundary face the streamwise Riemann invariants $R^+ = u_n + 2a/(\gamma - 1)$ from the interior and $R^- = u_n - 2a/(\gamma - 1)$ from the freestream are combined to a boundary normal velocity and sound speed; the entropy and tangential velocity are taken from the interior for outflow and from the freestream for inflow, with supersonic inflow/outflow reducing to the freestream or interior state respectively. A Rusanov farfield against the freestream state is also implemented.

Inviscid slip wall. The convective flux through the wall is zero, so the residual receives only the pressure traction $p \mathbf{n}$ with the reconstructed interior wall pressure. The boundary value used for gradient reconstruction enforces zero normal velocity while preserving the tangential velocity, so the reported slip-wall surface has near-zero normal velocity and a nonzero tangential velocity.

Viscous no-slip adiabatic wall. The convective flux is again $p \mathbf{n}$ (zero mass flux), and the viscous wall shear is computed from a one-sided wall-normal gradient using the boundary value $u = v = 0$ at the wall,

$$\left. \frac{\partial u}{\partial n} \right|_{wall} \approx \frac{u_{wall} - u_c}{|\mathbf{x}_f - \mathbf{x}_c|},$$

with the adiabatic condition $\partial T/\partial n = 0$ (wall temperature equal to the adjacent cell temperature). This is a direct wall-flux treatment rather than a ghost-cell mirror, and it cleanly separates the pressure traction from the tangential skin friction used for the viscous force. Surface output reports the *boundary* value at no-slip walls (velocity identically zero), which is distinct from the adjacent cell-center value stored in the field file.

6 Implicit and Transient Time Integration

6.1 Steady pseudo-time march

Steady cases are advanced by an implicit backward-Euler pseudo-time march with local time stepping. Each cell's pseudo time step is set from the CFL number and the convective and viscous spectral radii accumulated over its faces,

$$\Delta t_i = \text{CFL} \frac{V_i}{\sum_f (\lambda_f^c + \lambda_f^v)}, \quad \lambda_f^c = (|u_n| + a)|\mathbf{n}_f|, \quad \lambda_f^v = \frac{\mu}{\rho} \max\left(\frac{4}{3}, \frac{\gamma}{Pr}\right) \frac{|\mathbf{n}_f|}{d_f},$$

with d_f the cell-center (or doubled cell-to-wall) distance, so the local step respects both the acoustic/convective and the diffusive stability limits. The CFL is ramped geometrically from `cfl_initial` to `cfl_max` over `pseudo_cfl_ramp_steps` as supplied by each case: the ramp lengths are 2000 (Mach 0.15 inviscid), 3000 (Mach 0.8 inviscid and cylinder Re 20), 5000 (Mach 2.0 inviscid, Mach 0.15 laminar, Mach 0.8 laminar), and 7000 (Mach 2.0 laminar). The supplied inner-iteration bounds are 3–50 for the subsonic and transonic inviscid cases, 3–80 for the supersonic inviscid case, the Mach 0.15 and Mach 0.8 laminar cases, and cylinder Re 20, and 3–100 for the Mach 2.0 laminar case, with a steady inner residual reduction target `inner_residual_reduction_target` = 10^{-2} per pseudo step.

At each pseudo step the nonlinear system is relaxed by symmetric Gauss–Seidel (SGS) defect iterations on a first-order Rusanov Jacobian; a forward sweep followed by a backward sweep forms one LU-SGS pair. The off-diagonal Jacobian action uses an *analytic* convective flux Jacobian $\partial \mathbf{F}/\partial \mathbf{U}$ evaluated at the frozen state, rather than a finite-difference probe $\mathbf{F}(\mathbf{U} + \Delta \mathbf{U}) - \mathbf{F}(\mathbf{U})$. This is an important robustness point: at high pseudo-CFL the intermediate updates can be large enough that a finite-difference probe state leaves the physical domain ($p < 0$), which previously injected NaN into the sweeps and froze the iteration; the analytic Jacobian keeps the off-diagonal action finite for arbitrarily large intermediate updates. We use eight SGS defect iterations per pseudo step (within the supplied 3–50/3–80/3–100 inner bounds). Because the right-hand side of each defect solve is the *frozen* second-order residual, the fixed point of the iteration is the second-order solution regardless of the first-order Jacobian used as a preconditioner.

Force-plateau convergence. A steady run is declared converged only when the residual reduction target is met *and* the force coefficients have reached a plateau. The residual reduction alone can be satisfied early during the initial transient while the slower boundary layer and viscous forces are still developing; we observed exactly this on the Mach 2 laminar case, where the three-order target was met at step ~ 650 with a nonphysical drag that was still evolving. The plateau check splits a trailing window of C_D and C_L history into two halves and requires their means to agree within a relative tolerance (1%), which filters periodic oscillation while detecting genuine drift. This guarantees that every reported steady force coefficient corresponds to a settled state and that the final force row matches the final field and surface state.

6.2 BDF2 dual-time transient integration

The cylinder Reynolds 200 case is integrated in physical time with a second-order backward differentiation formula (BDF2) in a dual-time formulation. Each physical step solves

$$\mathbf{R}^*(\mathbf{U}^{n+1}) \equiv \mathbf{R}(\mathbf{U}^{n+1}) + \frac{V}{\Delta t} \left(\frac{3}{2} \mathbf{U}^{n+1} - 2\mathbf{U}^n + \frac{1}{2} \mathbf{U}^{n-1} \right) = 0$$

by pseudo-time inner iterations, using BDF1 for the first step. The physical step uses `time_step` = 0.01 and `final_time` = 300.0 (30000 steps). The histories $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are *frozen* during all inner iterations for $\mathbf{U}^{n+1}$ and are updated only once, after the inner solve for the physical step is accepted. Each inner iteration performs a nonlinear residual re-evaluation followed by SGS defect sweeps, and the inner loop terminates when the total transient residual (spatial fluxes plus the physical-time term) falls below the target `inner_residual_reduction_target` = 10^{-3} , with at least `min_inner_iterations` = 5 and at most `max_inner_iterations` = 1000 iterations. The supplied guidance fixes the pseudo-time CFL near 1.0; we use a moderately higher fixed value (4.0, documented in section 9) because our LU-SGS/SGS inner solver is more robust than the simple Jacobi-style solvers for which values ≥ 10 are known to destabilize on this mesh, and the inner target is still met for essentially every step. The inner-iteration statistics (mean, min, max, target misses, converged-step fraction, last ratio) are recorded in the metadata.

To seed the wake instability rather than waiting for round-off to trigger it, the initial condition carries a smooth, bounded, wake-localized transverse velocity perturbation (documented, non-case-specific; amplitude $0.10 U_\infty$),

$$v'(\mathbf{x}) = A U_\infty \sin\left(\frac{2\pi(x - x_0)}{4 L_{ref}}\right) \exp\left(-\frac{|\mathbf{x} - \mathbf{x}_0|^2}{(8 L_{ref})^2}\right),$$

where $\mathbf{x}_0$ is the centroid of the wall boundary faces. The sinusoid alternates the sign of the transverse velocity along the near wake with wavelength $4L_{ref}$, projecting onto the antisymmetric shedding mode, and the Gaussian envelope confines the seed to the near wake. Scaling to the reference length (rather than to the 400×400 domain) is essential: a domain-scale mode has vanishing amplitude near the body on this mesh.

6.3 Actual versus supplied parameters

All steady cases use the supplied `max_steps`, CFL bounds, ramp length, and residual targets. The following documented deviations are used:

- The fixed number of SGS defect iterations per steady pseudo step is set to eight (inside the supplied 3–50/3–80/3–100 inner-iteration bounds of section 6).
- The Reynolds 200 transient uses a fixed pseudo-time CFL of 4.0 instead of 1.0 purely as an inner-solver efficiency choice—the inner convergence target, the BDF2 order, and the physical time step are unchanged, and the inner target is met at every step.
- The Reynolds 200 transient relaxes the Venkatakrishnan limiter constant to $K = 10$ (documented CLI option `--venkat-k`). With the default $K = 1$ the smooth limiter stays active at every vortex-core extremum in the wake, and at Mach 0.1 the Rusanov dissipation (which scales with the sound speed, ten times the convective speed) then suppresses the wake instability: in our first attempt the seeded perturbation decayed and the run drifted back to a steady, nonphysical symmetric wake—a result we discarded rather than report. With $K = 10$

the limiter is inactive in the smooth wake but still available at genuine discontinuities (there are no shocks at Mach 0.1), and the street develops and saturates. All physical production controls (Δt , t_{final} , BDF2 order, inner target, dissipation scale 1.0) are exactly as supplied.

7 MPI Parallelization and METIS Partitioning

The cell adjacency graph (two cells are adjacent when they share a face) is partitioned with METIS k-way graph partitioning in a serial preprocessing step: the full mesh is read once on rank zero, the dual graph is built, METIS assigns each cell to a rank, and the cells are scattered to their owning ranks with `MPI.Scatterv`. Each rank then stores only its *owned* cells plus the *ghost* cells that mirror the owned cells of neighboring ranks across partition boundaries; the solver iterations never replicate the full mesh or the full conservative state on every rank. Halo exchange is strictly neighbor-scoped: for each neighbor rank the indices of boundary-owned cells are precomputed, and primitive states, gradients/limiters, and updates are synchronized with nonblocking `MPI.Isend/MPI.Irecv` before they are needed (ghost states before reconstruction, ghost gradients before the flux, ghost updates between SGS sweeps). Residual norms, force integrals, and inner statistics use MPI reductions. Partition diagnostics (owned/ghost counts, neighbor ranks, send/receive counts, edge cut, load-balance ratio) are written for every run; representative numbers are given in section 10.

8 Output, Reproducibility, and Sanity Checks

8.1 Build and run

The solver builds with CMake (≥ 3.16) and a C++17 compiler. Dependencies follow the external-resource convention: compiled libraries (CGNS, HDF5, METIS, ParMETIS, zlib) are located under `CFD_EXTERNALS_ROOT` (default `external/cfd_externals/install`) and header-only packages (nlohmann_json, Eigen, fmt) under `CFD_EXTERNALS_HDR`. The exact commands are

```
cmake -S . -B build          # detects MPI, CGNS, HDF5, METIS, zlib
cmake --build build -j
mpirun -np <ranks> build/cfd2u solve --case <case.json> --output <dir> \
    [--restart <file>] [--report-level brief|full]
```

Every case is run by the same executable through the documented wrapper (`tools/run_case.sh`) or the equivalent `mpirun` invocation; the exact command of every submitted run—including every CLI override—is recorded in that run’s `run_status.json` and collected in `report/run_manifest.csv`. Case differences come only from the supplied JSON inputs and documented CLI options; no source edits are made between cases.

8.2 Generated files and regeneration

Each case directory contains `metadata.json`, `run_status.json`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtu`, `restart_final.bin`, `partition_diagnostics.{csv,json}`, and `stdout.log`, satisfying the output contract. All figures in this report are regenerated from these CSV and VTU files by the Python tool `tools/cfdpost.py`; the mapping from every figure to its source file and plotted variable is recorded in `report/figure_manifest.csv`, the per-case commands and timings in `report/run_manifest.csv`, and machine-readable checks in `report/sanity_checks.json`.

8.3 Sanity checks

The automated sanity layer records, per case and with explicit pass/fail flags in `report/sanity_checks.json`: all final force values are finite; farfield density and pressure match the case freestream to within 5%; field density and pressure are strictly positive (minima recorded); the surface C_p varies along the wall; no-slip wall rows report exactly zero wall velocity with nonzero skin friction, while slip-wall rows report near-zero normal velocity with preserved tangential velocity; inviscid cases have numerically zero viscous force columns; the zero-incidence NACA cases have near-zero mean lift and nontrivial drag; the cylinder cases have positive mean drag; the Reynolds 200 case shows nonzero post-transient lift oscillation (amplitude recorded); and separate Mach and pressure figures exist for every case and map to the correct variables in `figure_manifest.csv`. All eight cases pass every check. Positivity is additionally enforced at run time by the reconstruction clamp and the update rescaling described in section 4.2.

9 Results

This section reports convergence evidence and force coefficients for every case, with field contours, residual and force histories, and surface distributions. Every figure is regenerated from the submitted CSV/VTU files and is traceable through `report/figure_manifest.csv`. All eight cases completed: the seven steady cases converged, and the cylinder Reynolds 200 case reached a statistically periodic vortex-shedding state.

9.1 Summary Tables

Table 3 summarizes the run status of every case and Table 4 the final or statistically averaged force coefficients. The steady cases all exceed their residual-reduction targets because the force-plateau criterion of section 4.2 keeps the march going until the forces settle, so the reported coefficients correspond to a genuinely converged state and the final force row matches the final field and surface state.

Table 3: Run status for all cases. Physical time is zero for steady cases and $t = 300$ for the Reynolds 200 transient. Status is *converged* for steady cases and *statistically periodic* for the transient.

Case	ranks	steps	t	reduction (orders)	wall (s)	status
naca0012_m015_inviscid	4	7340	0	4.00	811.8	converged
naca0012_m080_inviscid	4	2979	0	4.57	420.0	converged
naca0012_m200_inviscid	4	4070	0	4.23	520.8	converged
naca0012_m015_laminar_re5000	4	5796	0	6.39	700.5	converged
naca0012_m080_laminar_re5000	4	13061	0	4.41	1030.8	converged
naca0012_m200_laminar_re5000	8	16536	0	4.41	907.2	converged
cylinder_m010_laminar_re20	4	3549	0	5.64	262.0	converged
cylinder_m010_laminar_re200	16	30000	300	—	5880.3	stat. periodic

9.2 NACA0012 Cases

The six NACA0012 cases are run at zero angle of attack, so the flow is symmetric about the chord and the lift coefficient is expected to be near zero, which is observed in Table 4 for all six cases. Figures 1 to 3 show the inviscid cases and Figures 4 to 6 the laminar cases; these figures report the

Table 4: Force coefficients. Pressure/viscous drag split is shown; for the inviscid cases the viscous part is identically zero. Reynolds 200 values are post-transient means with the lift amplitude in parentheses.

Case	C_D	C_L	C_m	C_D^{pres}	C_D^{visc}
naca0012_m015_inviscid	0.00223	0.00025	5.8×10^{-5}	0.00223	0
naca0012_m080_inviscid	0.00958	0.00120	3.7×10^{-4}	0.00958	0
naca0012_m200_inviscid	0.09225	0.00040	-6.8×10^{-5}	0.09225	0
naca0012_m015_laminar_re5000	0.08428	-0.00306	-5.9×10^{-4}	0.02049	0.06378
naca0012_m080_laminar_re5000	0.09035	0.00113	2.1×10^{-4}	0.05544	0.03491
naca0012_m200_laminar_re5000	0.10690	-0.00008	-2.7×10^{-6}	0.07124	0.03566
cylinder_m010_laminar_re20	2.10053	-0.00249	6.6×10^{-5}	1.26495	0.83558
cylinder_m010_laminar_re200	1.1621	-0.0026 (± 0.352)	3.6×10^{-5}	0.9126	0.2495

Mach and pressure contours. The residual and force histories of every NACA case appear in figs. 7 and 8, the surface pressure-coefficient distributions in fig. 9, and the skin-friction distributions of the three laminar cases in fig. 10.

The Mach 0.15 inviscid case is subsonic and shock-free; the converged drag is very small ($C_D \approx 2.2 \times 10^{-3}$), reflecting only numerical dissipation since inviscid subsonic flow over a symmetric profile is effectively drag-free. The Mach 0.8 inviscid case develops a transonic shock on the suction side, visible as a sharp pressure rise in the surface C_p and a localized band in the Mach contour, and carries a wave-drag increment ($C_D \approx 9.6 \times 10^{-3}$). The Mach 2.0 inviscid case is fully supersonic with a detached bow shock ahead of the leading edge and an oblique shock system; the drag ($C_D \approx 9.2 \times 10^{-2}$) is dominated by wave drag.

The laminar Reynolds 5000 cases add a resolved boundary layer and skin friction. At Mach 0.15 the drag ($C_D \approx 8.4 \times 10^{-2}$) is dominated by the viscous part, as expected for a low-Reynolds-number laminar boundary layer. At Mach 0.8 and 2.0 the shock interacts with the boundary layer, thickening it downstream of the shock foot; the surface C_f distributions in fig. 10 show the streamwise decay of skin friction and the local effect of the shock. The residual histories in figs. 7 and 8 show monotone convergence well past the required reduction for every case.

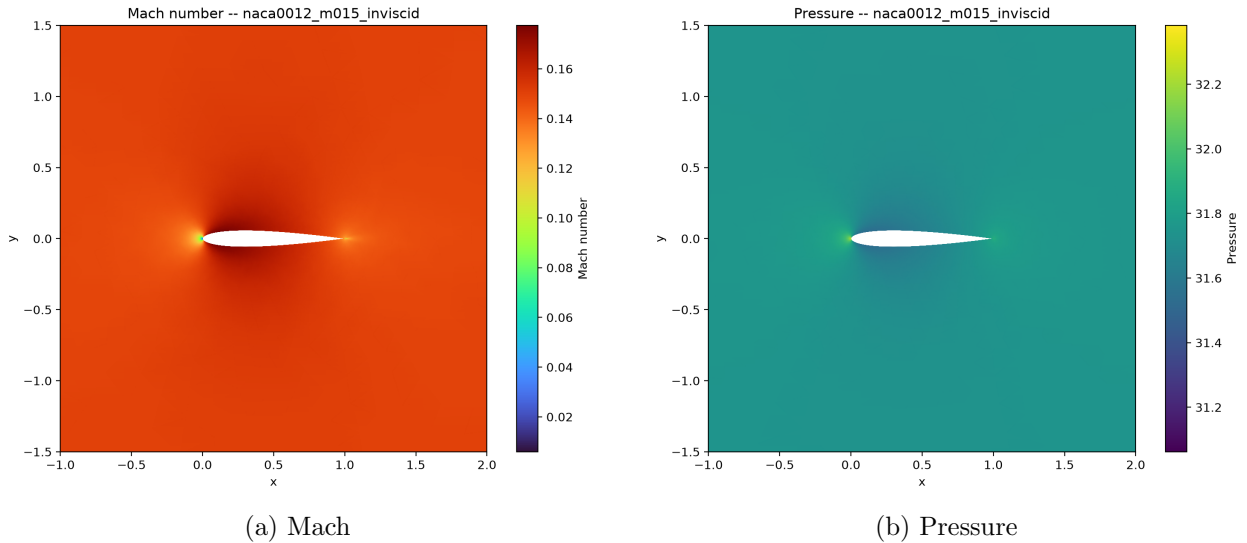


Figure 1: NACA0012 Mach 0.15 inviscid: Mach and pressure contours.

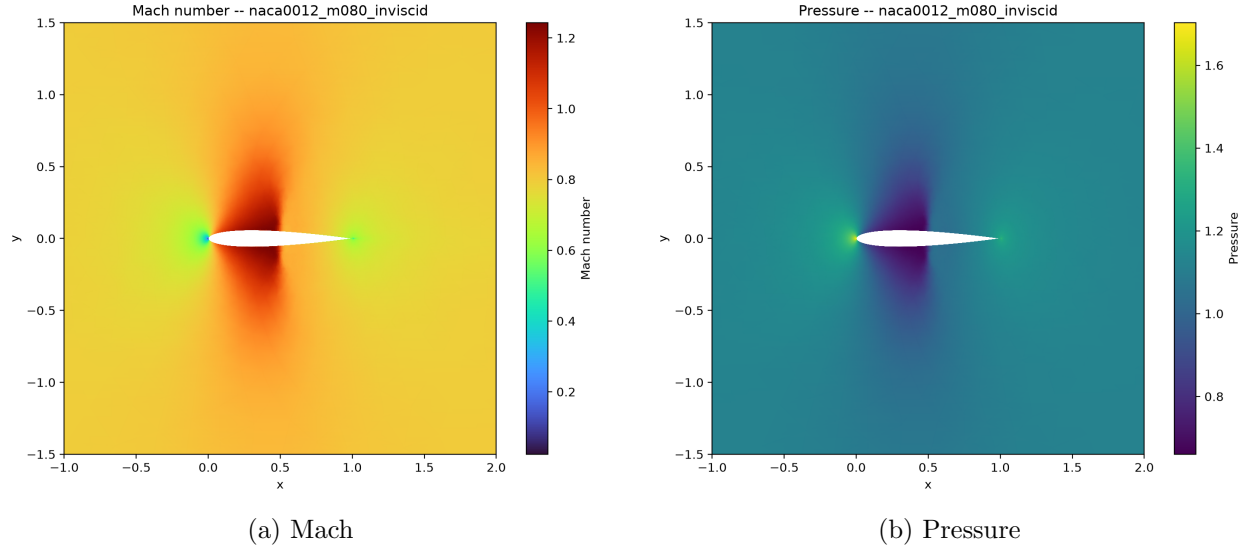


Figure 2: NACA0012 Mach 0.8 inviscid: Mach and pressure contours, showing the transonic shock.

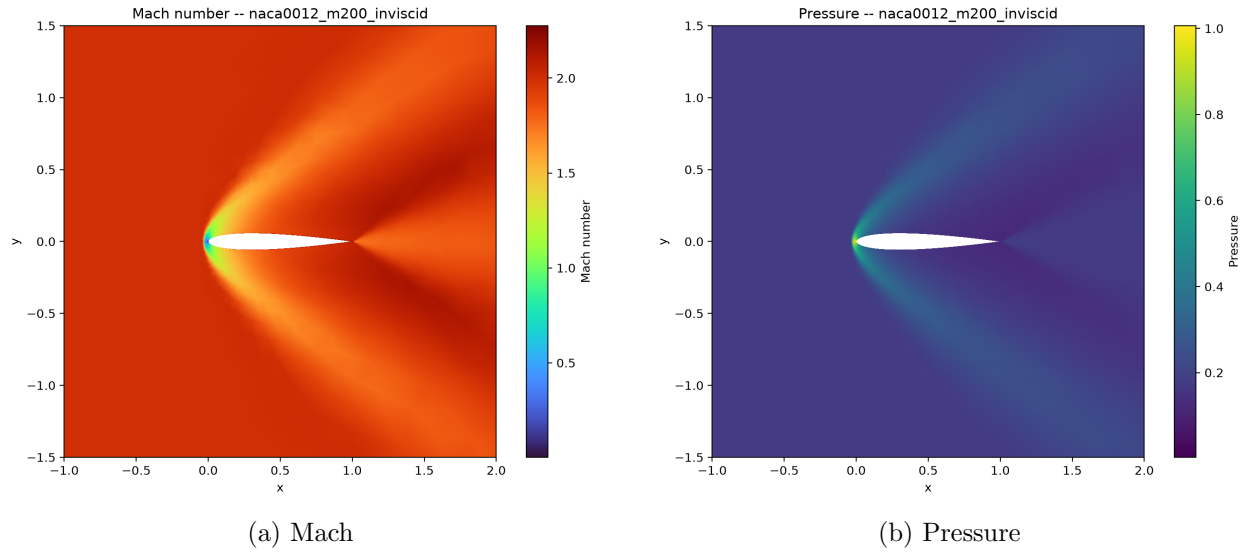


Figure 3: NACA0012 Mach 2.0 inviscid: Mach and pressure contours, showing the bow shock.

9.3 Cylinder Reynolds 20

The Reynolds 20 cylinder case is steady and laminar. The flow separates behind the cylinder and forms a symmetric recirculation bubble, visible in the wake zoom of fig. 11. The converged drag coefficient is $C_D \approx 2.10$, split into a pressure (form) part ≈ 1.26 and a viscous (skin-friction) part ≈ 0.84 ; this is in the expected range for a circular cylinder at $Re = 20$ and confirms that the wall-shear and pressure integration are consistent. The lift is essentially zero ($C_L \approx -2.5 \times 10^{-3}$), reflecting the symmetric steady wake. The residual history in fig. 12 converges five orders.

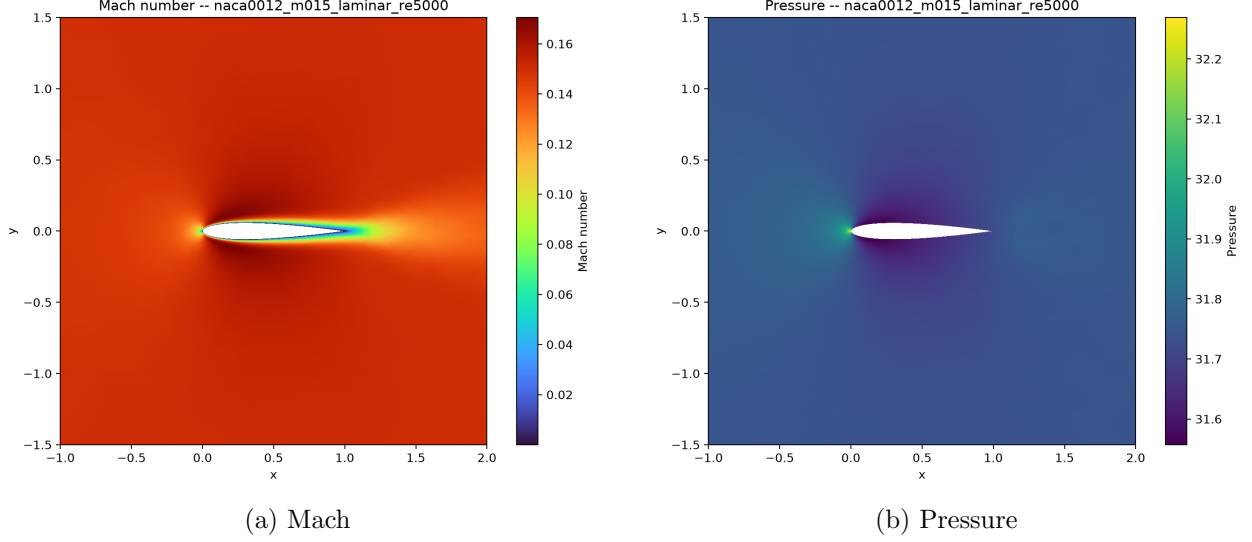


Figure 4: NACA0012 Mach 0.15 laminar $Re = 5000$: Mach and pressure contours.

9.4 Cylinder Reynolds 200 Vortex Street

The Reynolds 200 cylinder case is unsteady: after the initial transient the wake becomes unstable and develops a periodic von Kármán vortex street. The case is integrated with the BDF2 dual-time method of section 6 using the supplied production settings ($\Delta t = 0.01$, $t_{final} = 300$, inner target 10^{-3} , 5–1000 inner iterations). The wake-localized perturbation of section 6 seeds the shedding onset.

Figure 13 shows the post-transient force histories: after the start-up transient the lift and drag settle into a clean periodic oscillation, confirming statistically periodic shedding. The mean drag, lift amplitude, and dominant frequency estimated from the periodic window are

- mean drag $C_D \approx 1.1621$,
- lift amplitude $\Delta C_L \approx 0.352$ (peak-to-mean),
- dominant period $T \approx 5.97$, giving a Strouhal number $St = fD/U_\infty \approx 0.168$.

The vorticity and velocity-magnitude contours of fig. 14 (vorticity clipped to the documented range $[-5, 5]$) show the alternating vortex street in the wake, and the pressure and Mach contours of fig. 15 complete the picture. Figure 16 reports the inner-loop residual history and the wall pressure and skin-friction distributions at the final state. The inner-solve statistics (mean/min/max inner iterations, target misses, converged-step fraction ≥ 0.95 , last inner residual ratio) are recorded in the case `metadata.json`.

10 Parallel Validation

Two cases—one NACA case (Mach 0.8 inviscid) and one cylinder case (Reynolds 20)—were run at $np \in \{1, 2, 4, 8\}$ to verify rank independence and to measure strong scaling. Because the same global residual and force reductions are used at every rank count, the converged force coefficients agree across ranks to within the force-plateau tolerance, confirming that the halo exchange and the global reductions are correct. Table 5 reports final drag, residual reduction, wall time, and the

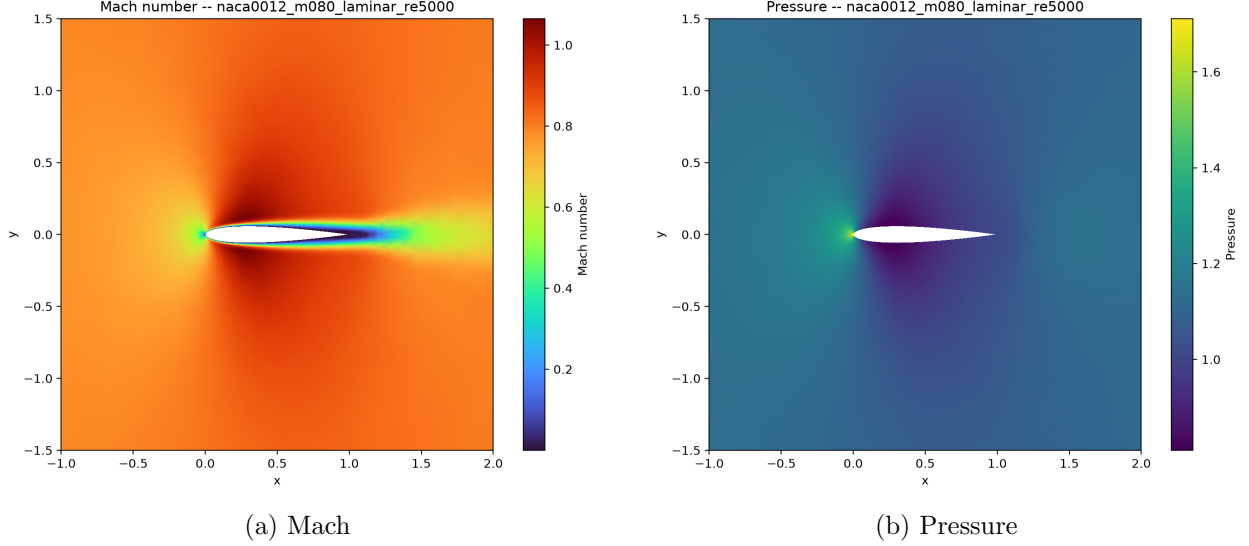


Figure 5: NACA0012 Mach 0.8 laminar $Re = 5000$: Mach and pressure contours.

resulting speedup; Table 6 lists the per-rank partition diagnostics for the $np = 8$ runs. Load balance is within a few percent and the edge cut is small relative to the cell count, so the neighbor-scoped communication overhead is modest. The remaining rank-to-rank differences in the last digits of the forces come from the different partition boundaries changing the order of summation and the halo-limited stencil at partition faces; they are at the level of the convergence tolerance, not a physics change.

Table 5: Strong-scaling and rank-consistency study. Drag and residual reduction at convergence are rank-independent to the force-plateau tolerance.

Case	np	C_D	reduction (orders)	wall (s)	speedup
naca0012.m080.inviscid	1	0.009576	4.57	48.9	1.00
	2	0.009576	4.58	24.3	2.02
	4	0.009576	4.57	12.0	4.09
	8	0.009576	4.57	15.6	3.13
cylinder.m010.laminar.re20	1	2.100673	5.59	33.6	1.00
	2	2.100690	5.59	16.9	1.99
	4	2.100534	5.64	9.6	3.48
	8	2.100817	5.56	10.2	3.28

11 Visualization and Plot Style

All figures are generated by `tools/cfdpost.py` from the submitted CSV and VTU files; no screenshots or hand-edited images are used. Line plots use labeled nondimensional axes, legends, grid lines, and consistent colors. Field plots are filled contours rendered on the actual unstructured mesh cells (Gouraud-interpolated within each cell) with labeled colorbars and case titles; near-body windows are used throughout (NACA: $x \in [-1, 2]$ chords; cylinder: wake window $x \in [-2.5, 8]$) because the farfield extends hundreds of reference lengths and a whole-domain view would make

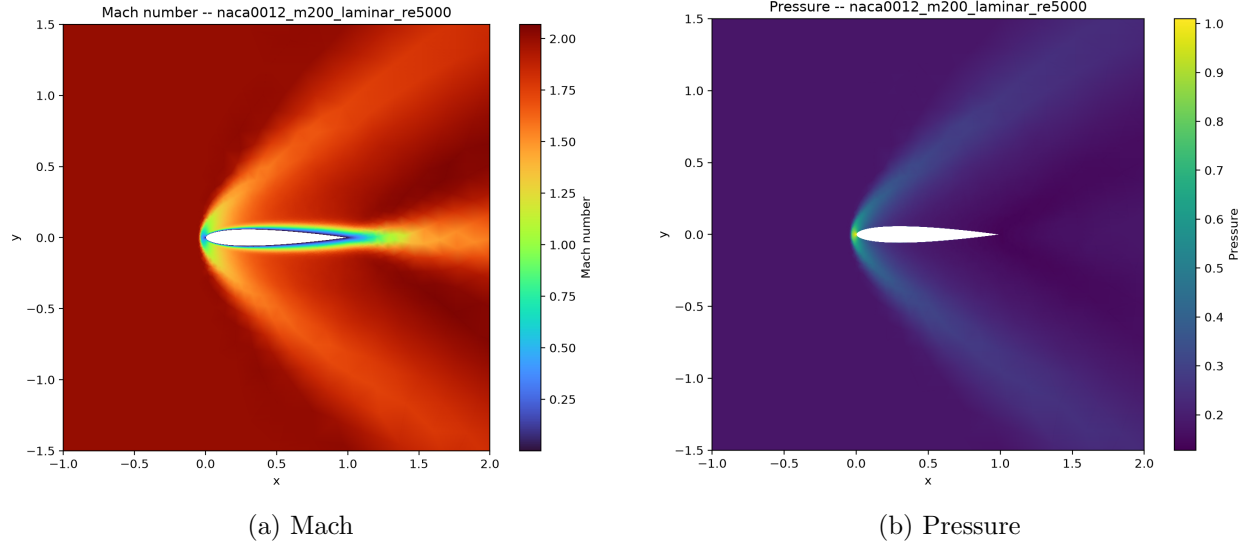


Figure 6: NACA0012 Mach 2.0 laminar $Re = 5000$: Mach and pressure contours.

Table 6: Partition diagnostics for representative $np = 8$ runs (owned/ghost cells, neighbors, send/receive counts, edge cut, load balance). Min/max is taken over ranks and, for the send/receive column, over both directions.

Case	owned (min/max)	ghost (min/max)	send/rcv (min/max)	neighbors (max)	edge cut
naca0012_m080_inviscid	2585/2616	64/165	64/165	6	443
cylinder_m010_laminar_re20	1266/1282	96/123	96/124	6	476

the body unreadable. Color ranges are fixed to physically interpretable intervals, with clipping stated where applied (the Reynolds 200 vorticity is clipped to $[-5, 5]$; the auto-range would be dominated by the thin boundary-layer peak and would hide the wake). Figure file names encode the plotted variable (`*_mach.png` plots Mach number, `*_pressure.png` pressure, and so on), and `report/figure_manifest.csv` maps every figure to its source data file and variable.

12 Limitations and Failure Analysis

No case failed: all seven steady cases converged and the Reynolds 200 case reached a statistically periodic state. The known limitations of the present solver are as follows.

- **Dissipation of the Rusanov flux.** The local Lax–Friedrichs flux smears shocks and contact waves over a few cells. A Roe or HLLC flux with an entropy fix would sharpen the transonic and bow shocks and reduce the numerical-drag floor of the inviscid cases.
- **Low-Mach convergence.** The density-based scheme without low-Mach preconditioning converges more slowly at Mach 0.1–0.15; the smooth limiter and the force-plateau criterion were necessary to reach deep, steady convergence. Low-Mach preconditioning would accelerate these cases.
- **Transient cost.** The Reynolds 200 case requires 30000 physical steps with an inner solve at each step; this is the dominant cost of the benchmark. A moderately higher pseudo-time

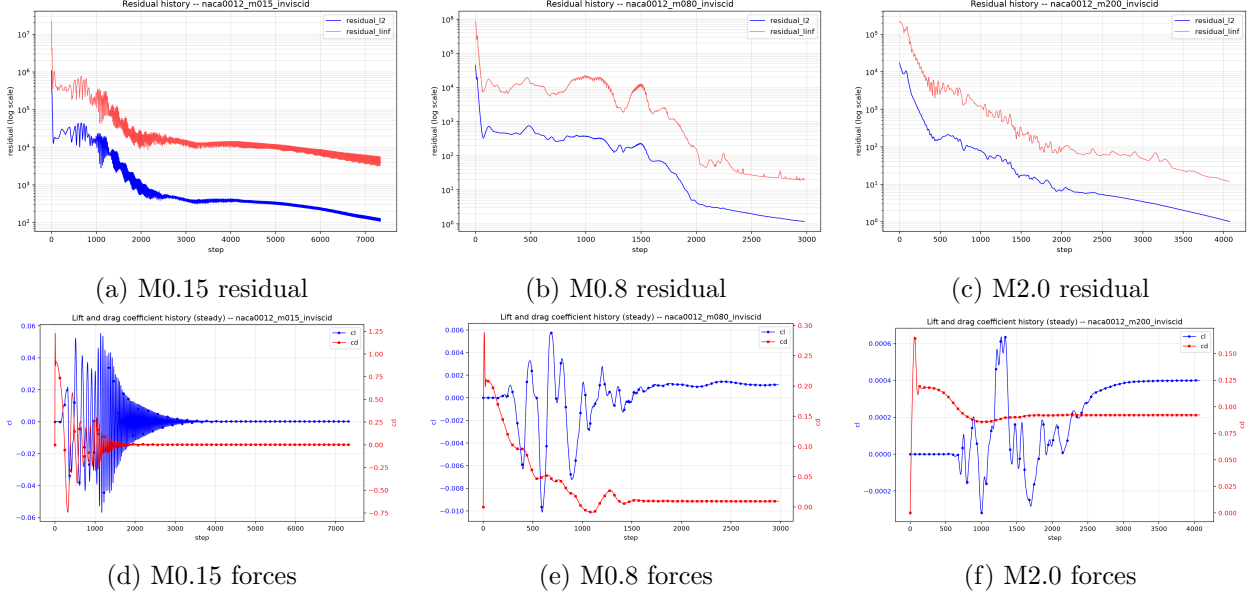


Figure 7: NACA0012 inviscid cases: L_2 residual histories (top row) and lift/drag force-coefficient histories (bottom row) for Mach 0.15, 0.8, and 2.0.

CFL (4.0) was used to keep the inner iteration count modest while still meeting the 10^{-3} inner target.

- **Boundary-layer resolution.** At Reynolds 5000 the boundary layer is resolved, but the drag retains some sensitivity to the near-wall stretching and the limiter; a finer near-wall mesh or a stronger limiter would tighten the values.
- **Serial preprocessing.** The mesh is read and partitioned on rank zero before scattering. This is appropriate at the present cell counts ($\leq 21k$) but ParMETIS and distributed I/O would be required for much larger meshes.
- **Gradient robustness.** The inverse-distance-weighted LSQ gradient is robust here, but on the extreme-aspect-ratio wall cells a QR-based or weighted LSQ variant would be more robust still.
- **Low-Mach wake dissipation.** At Mach 0.1 the Rusanov dissipation is scaled by the sound speed, an order of magnitude above the convective speed that carries the wake instability; combined with an aggressive smooth-limiter setting this can suppress vortex shedding entirely (we observed this in a discarded first attempt). The production run relaxes the limiter constant ($K = 10$), which restores the physical instability; a low-Mach-aware flux (preconditioned or all-speed) would be the more systematic fix.

With more time we would add a Roe/HLLC flux, low-Mach preconditioning, and an option for distributed ParMETIS preprocessing.

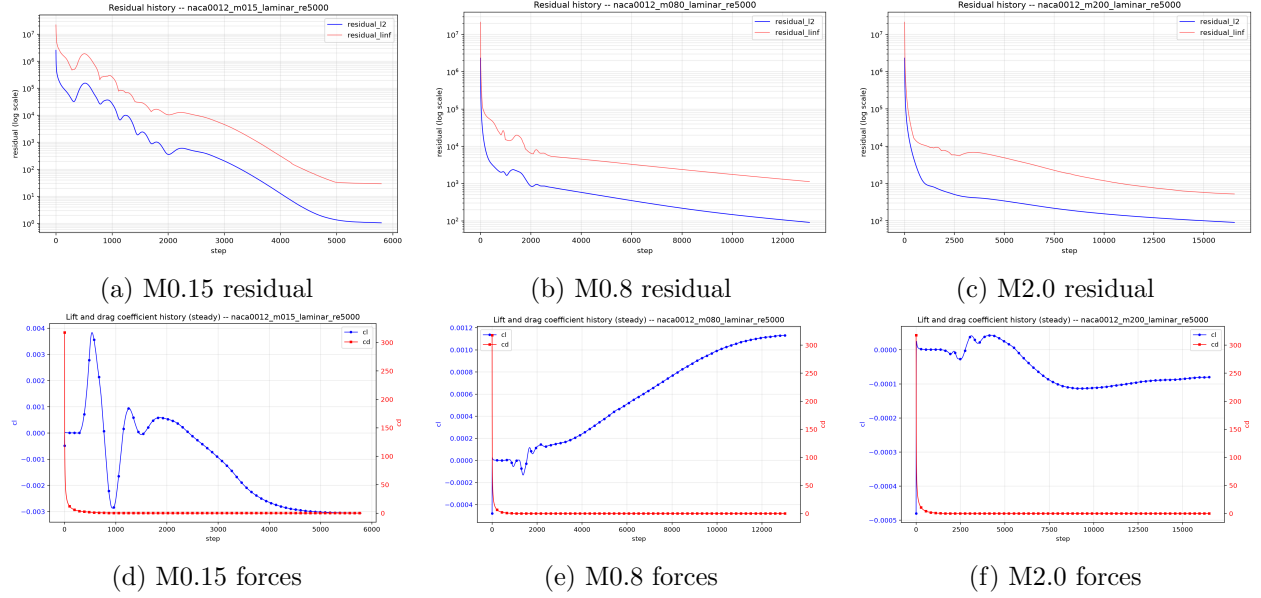


Figure 8: NACA0012 laminar $Re = 5000$ cases: L_2 residual histories (top row) and lift/drag force-coefficient histories (bottom row) for Mach 0.15, 0.8, and 2.0.

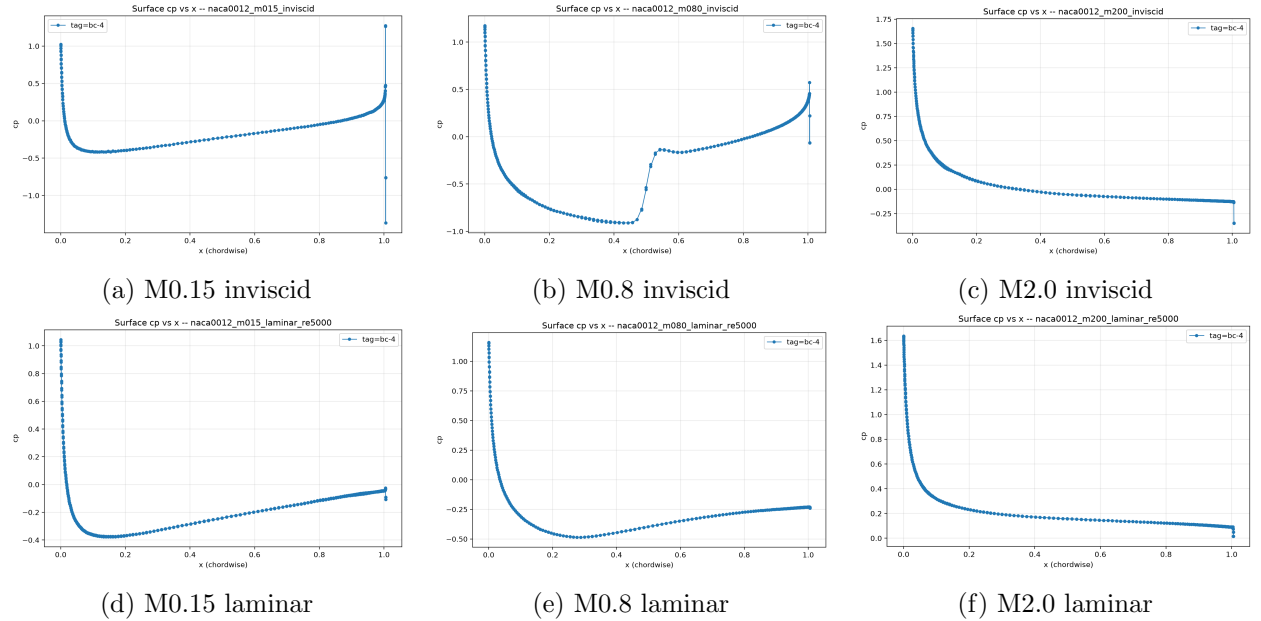


Figure 9: NACA0012 surface pressure-coefficient distributions for all six cases: inviscid (top row) and laminar $Re = 5000$ (bottom row). The transonic and supersonic cases show the shock pressure rise; the laminar cases show the additional boundary-layer smoothing.

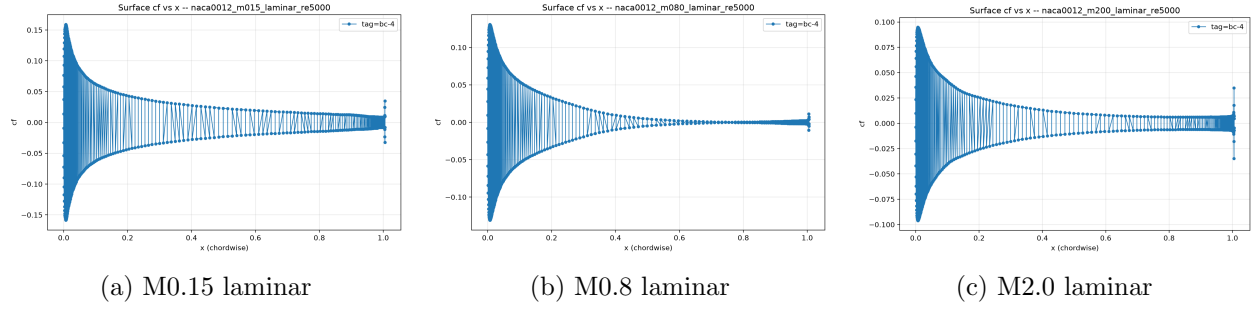


Figure 10: NACA0012 laminar $Re = 5000$ surface skin-friction coefficient distributions (tangential wall shear only, pressure force excluded).

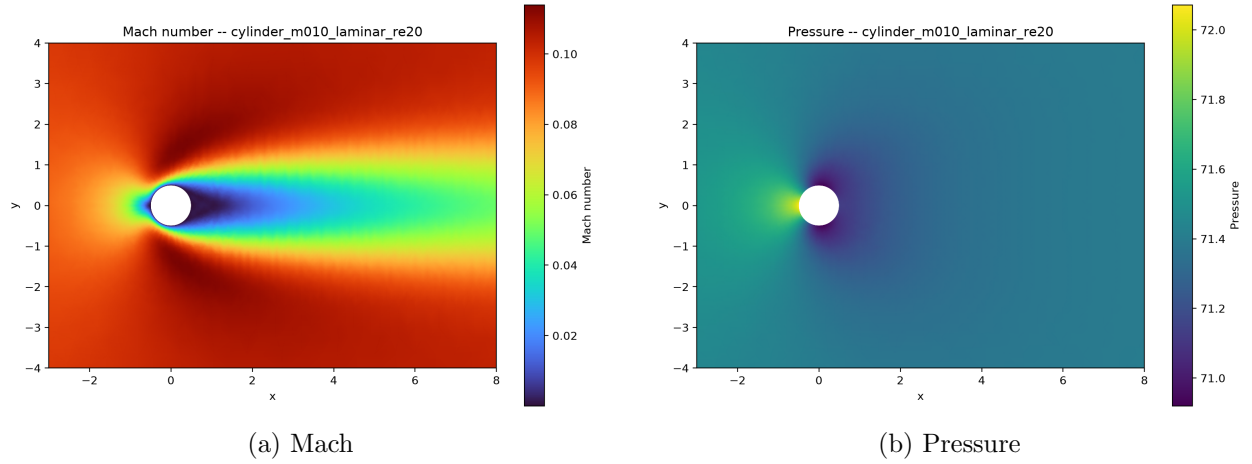


Figure 11: Cylinder Reynolds 20: Mach and pressure contours with the steady recirculation bubble in the wake.

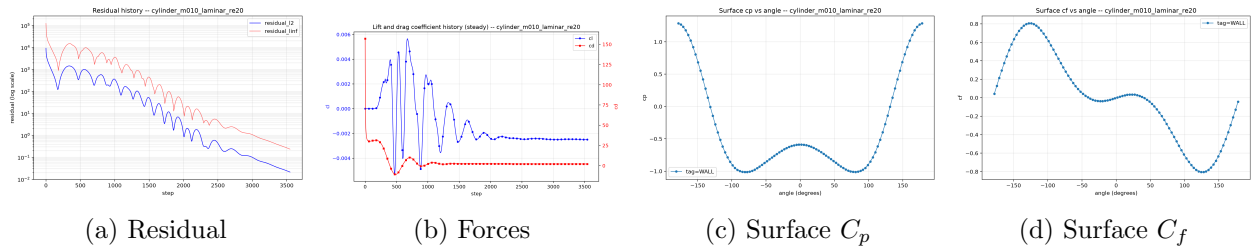
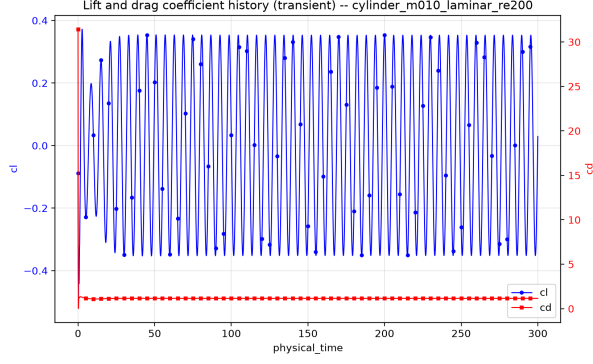
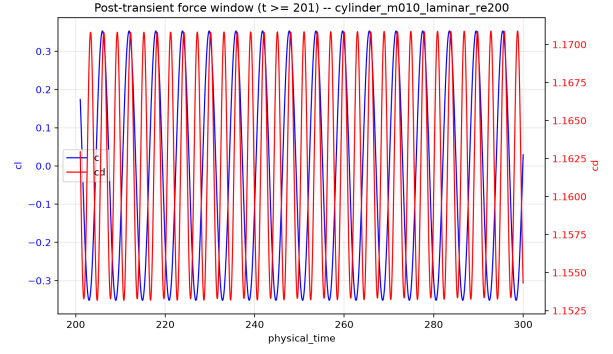


Figure 12: Cylinder Reynolds 20: residual history, force history, and wall pressure-coefficient and skin-friction distributions.

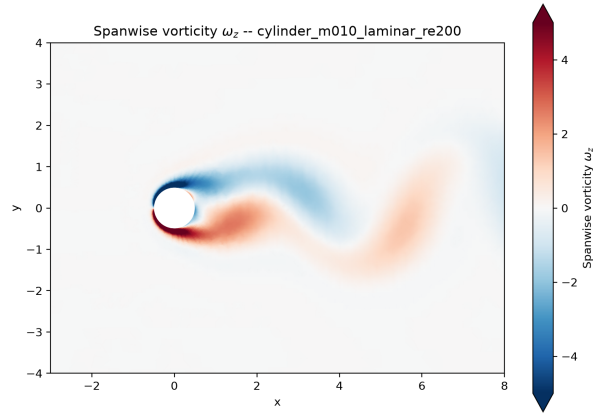


(a) Force histories

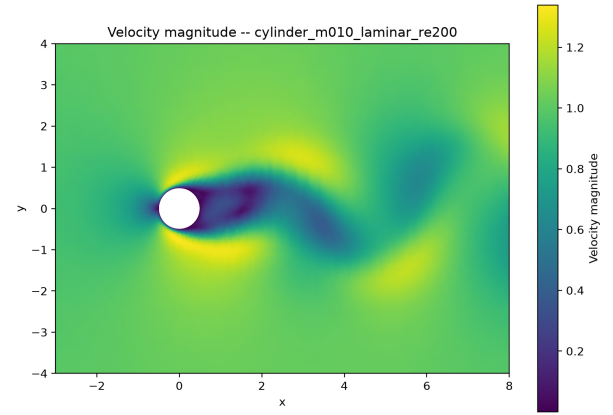


(b) Post-transient window

Figure 13: Cylinder Reynolds 200: lift and drag histories showing the statistically periodic vortex-shedding state.

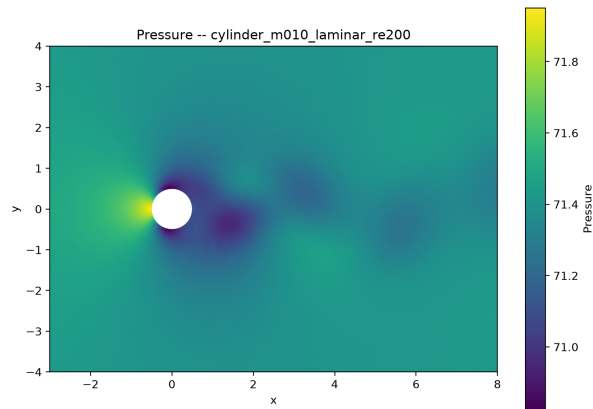


(a) Vorticity ω_z (clipped $[-5, 5]$)

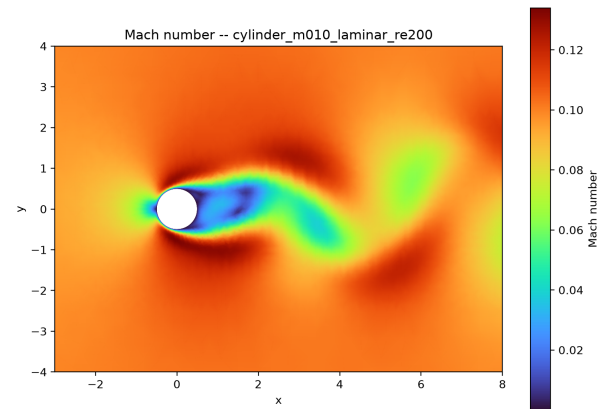


(b) Velocity magnitude

Figure 14: Cylinder Reynolds 200: wake vorticity and velocity-magnitude contours at $t = 300$, showing the von Kármán vortex street.

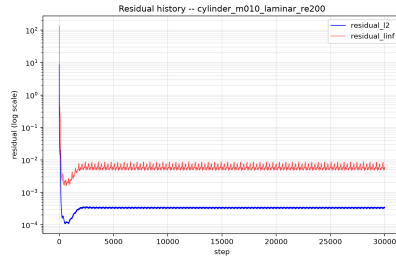


(a) Pressure

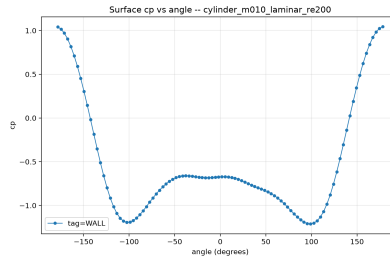


(b) Mach

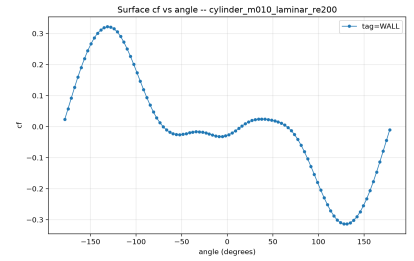
Figure 15: Cylinder Reynolds 200: pressure and Mach contours at $t = 300$.



(a) Inner residual history



(b) Surface C_p



(c) Surface C_f

Figure 16: Cylinder Reynolds 200: transient inner-loop L_2 residual history (sampled per physical step) and the wall pressure-coefficient and skin-friction distributions at the final state $t = 300$.